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Roll No. .....

324314(24)

B.E. (Third Semester)

Examination, April/May 2015

(Branch : Elect., EEE)

ELECTRIC CIRCUITS

Time : Three Hours] *[Maximum Marks : 80*

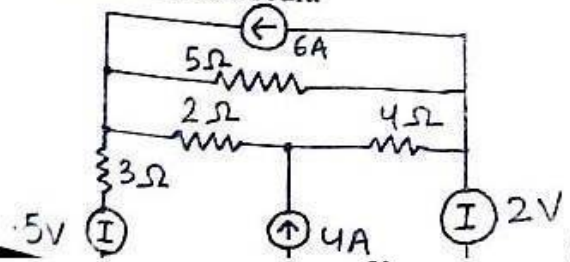
[Minimum Pass Marks : 28

Note : Part (a) of each question is compulsory. Attempt any two parts from (b), (c) and (d).

1. (a) Define twigs and links in network graph theory. 2

(b) Explain the method of obtaining loop matrix from a tree with suitable examples. 7

(c) Using nodal method find the current through the resistors in the circuit. 7



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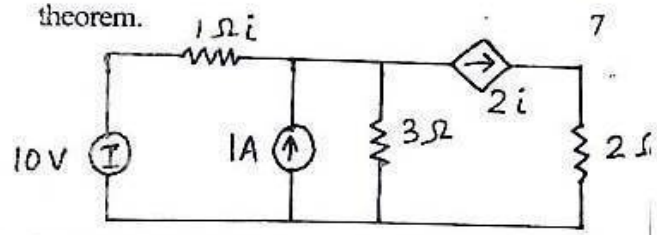
(d) Draw the graph and hence obtain the fundamental cut-set matrix of the given reduced incidence matrix : 7

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 & 1 & 0 & -1 \\ 0 & 1 & 0 & 0 & -1 & 1 & 1 \\ -1 & 0 & -1 & 0 & 0 & -1 & 0 \end{bmatrix}$$

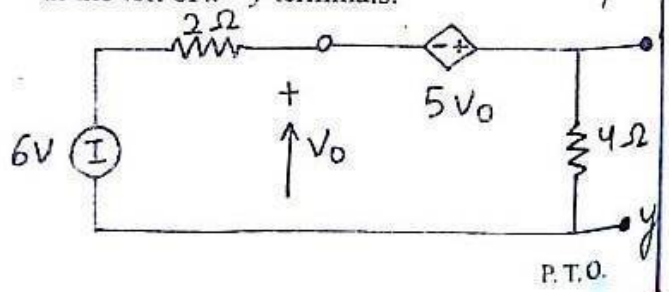
2. (a) Explain reciprocity theorem. 2

(b) Find the current in 2 Ω resistor using Thevenin's theorem. 7



(c) State and prove superposition theorem. 7

(d) Find the Norton's equivalent circuit shown in fig at the left of $x - y$ terminals. 7



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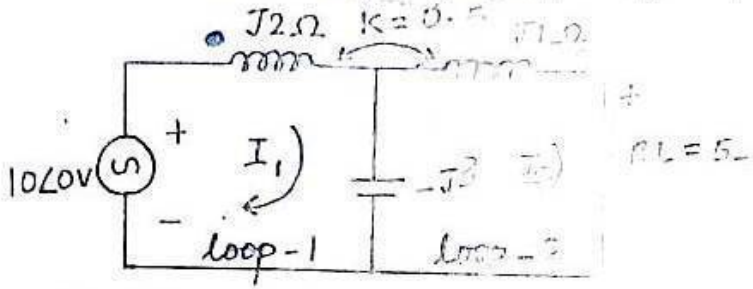
3. (a) Why we cannot change current across inductor instantaneously? 2

(b) Find the drop across R_1 in the given fig. 7



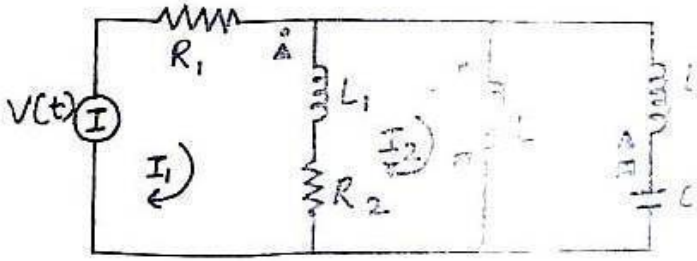
(c) Write short notes on the following: 7

(i) series and parallel resonance

(ii) quality factor

(iii) bandwidth

(d) Use a double subscript notation for mutual inductance write the Kirchhoff's voltage equations.



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4. (a) What is polyphase system ? 2

(b) Write down the relationship between line voltage and line current with phase voltage and phase current in star connected and delta connected load. 7

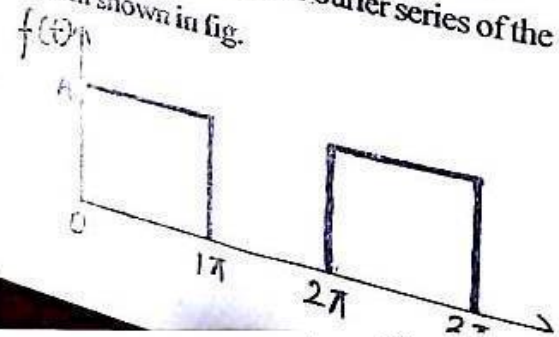
(c) A three phase balanced system supplies 110 V to a delta connected load whose phase impedences are equal to (3·54 + J 3·54) ohm. Determine the line current and draw the phasor diagram. 7

(d) Explain two-wattmeter method to measure power in a balanced 3-phase load. 7

5. (a) Define Fourier coefficients. 7

(b) Explain even, odd and half waveform symmetry. 2

(c) Find the trigonometric Fourier series of the wave form shown in fig. 7



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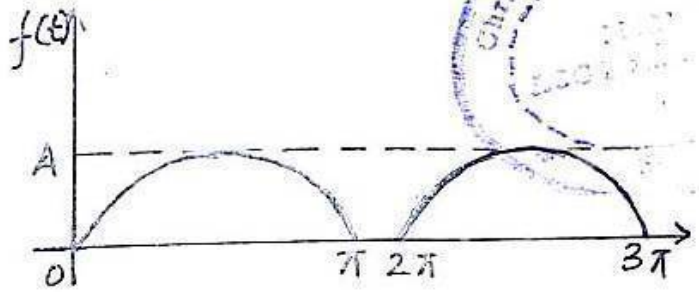
(d) Find the exponential fourier series of the waveform shown in fig. 7



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